

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: tsplib_failure_replay_compression.
- Best TSPLIB holdout gap: tsplib_failure_replay_compression.
- Best synthetic holdout gap: tsplib_failure_replay.

Run Metadata

- run_name: run_20260513_155249_t
- started_at_local: 2026-05-13 15:52:49
- finished_at_local: 2026-05-13 16:03:00
- duration_hhmm: 00:10
- duration_seconds: 611.12
- seed_offset: 19000
- replicate_label: t
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
tsplib_no_replay	none	score_only	False	0.213387	0.064916	0.159398	0.90376	0.56	-0.027557
tsplib_random_replay	random	score_only	False	0.150744	0.010343	0.099689	0.74996	0.76	0.001182
tsplib_failure_replay	failure	score_only	False	0.162444	0.0	0.103373	0.750785	0.76	0.015459
tsplib_failure_replay_compression	failure	novelty_gate	True	0.127075	0.0	0.080866	0.672957	0.76	0.012927

Condition Notes

tsplib_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.

- Accepted-epoch count 3, mean accepted code novelty 0.90376, and final complexity 0.56.
- Adaptation efficiency -0.027557 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.150744, synthetic 0.010343, combined 0.099689.
- Accepted-epoch count 3, mean accepted code novelty 0.74996, and final complexity 0.76.
- Adaptation efficiency 0.001182 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.150744 across 7 instances; family means: ch=0.145226, kroD=0.199962, pcb=0.264524, pr=0.109089, rd=0.145259, st=0.045926.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3408, "family": "a", "name": "a280", "optimality_gap": 0.321442}, {"best_known_cost": 7542, "cost": 9545, "family": "berlin", "name": "berlin52", "optimality_gap": 0.265579}, {"best_known_cost": 14379, "cost": 17716, "family": "lin", "name": "lin105", "optimality_gap": 0.232075}]

tsplib_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.162444, synthetic 0.0, combined 0.103373.
- Accepted-epoch count 3, mean accepted code novelty 0.750785, and final complexity 0.76.
- Adaptation efficiency 0.015459 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.162444 across 7 instances; family means: ch=0.077811, kroD=0.194045, pcb=0.286699, pr=0.298006, rd=0.174589, st=0.028148.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3465, "family": "a", "name": "a280", "optimality_gap": 0.343544}, {"best_known_cost": 14379, "cost": 17789, "family": "lin", "name": "lin105", "optimality_gap": 0.237151}, {"best_known_cost": 7542, "cost": 9051, "family": "berlin", "name": "berlin52", "optimality_gap": 0.20008}]

tsplib_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.127075, synthetic 0.0, combined 0.080866.
- Accepted-epoch count 3, mean accepted code novelty 0.672957, and final complexity 0.76.

- Adaptation efficiency 0.012927 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.127075 across 7 instances; family means: ch=0.100193, kroD=0.05147, pcb=0.189826, pr=0.301168, rd=0.121492, st=0.025185.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3327, "family": "a", "name": "a280", "optimality_gap": 0.290035}, {"best_known_cost": 14379, "cost": 15748, "family": "lin", "name": "lin105", "optimality_gap": 0.095208}, {"best_known_cost": 7542, "cost": 7964, "family": "berlin", "name": "berlin52", "optimality_gap": 0.055953}]

Judge Appendix

Interpretation of Results for Replay-Aware TSP Benchmark Suite

Condition	Synth etic Gap	TSP LIB Gap	Tran sfer Gap	Replay Mode	Code Novelty (mean)	Co mpl exit y	Notes on Transfer and Novelty
tsplib_no_replay	0.064916	0.213387	0.159398	none	0.90376	0.56	Worst transfer and TSPLIB gaps; highest code novelty but lowest transfer performance.
tsplib_random_replay	0.010343	0.150744	0.099689	random	0.74996	0.76	Improved transfer and synthetic gaps vs no replay; code novelty reduced compared to no replay.
tsplib_failure_replay	0.0	0.162444	0.103373	failure	0.75079	0.76	Synthetic gap minimized; TSPLIB gap improved over no replay but worse than random replay; code novelty decreased.
tsplib_failure_replay_compression	0.0	0.127075	0.080866	failure + compression	0.67296	0.76	Best TSPLIB and transfer gaps; synthetic gap zero; lowest code novelty and complexity controlled by compression.

Key Observations

- **Optimality Gap & Transfer Performance****
 - The ****tsplib_failure_replay_compression**** condition exhibits the best performance on held-out TSPLIB (mean gap 0.127) and transfer gap (0.081), indicating superior generalization and adaptation.
 - Failure replay conditions (with and without compression) outperform random replay and no replay in transfer metrics, with compression further enhancing transfer.
 - No replay condition yields the poorest transfer and TSPLIB optimality gaps despite maintaining the highest code novelty.
- **Code Novelty vs Transfer****
 - Code novelty decreases from no replay (0.90+) to failure replay (0.75) and further to failure replay with compression (0.67).

- Transfer performance improves as code novelty decreases, indicating that reduced novelty due to replay and compression is associated with improved optimality and transfer.
- This dissociation suggests that lexical novelty alone does not imply algorithmic invention; improved transfer is correlated with replay modes rather than novelty per se.

3. ****Replay Modes****

- No replay (baseline): Worst transfer and TSPLIB generalization gaps.
- Random replay: Lower gaps than no replay but not best overall.
- Failure replay: Further improvements relative to random replay.
- Failure replay + compression: Best transfer and TSPLIB gaps, achieving zero synthetic gap.
- Compression-aware replay reduces complexity (fixed at 0.76) compared to no replay (0.56 but with worse transfer), indicating effective control of solution complexity alongside transfer gains.

Summary

- ****Failure replay with compression pressure is the best transfer condition, achieving the lowest mean optimality gaps on held-out TSPLIB and synthetic sets.****
- ****This improvement coincides with a reduction in code novelty compared to no replay, demonstrating that enhanced transfer is not driven by increased lexical novelty.****
- ****Random replay provides moderate improvement over no replay but is outperformed by failure replay modes.****
- ****Replay strategies focusing on failure cases and compression are effective for transfer, confirming the value of replay-aware learning in TSP generalization.****